

RESEARCH ARTICLE

Defect-Enhanced Low-Temperature CO Oxidation on Graphene-Supported Pt-Au Nanoclusters

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ABSTRACT

The catalytic properties of bimetallic nanoclusters are governed by a complex interplay between their size, composition, chemical ordering, and interaction with the support. Here, we perform a comprehensive data-driven computational study using the USPEX evolutionary algorithm for global structure prediction of Pt-Au nanoclusters (up to 24 atoms), followed by high-throughput screening for O and CO adsorption energy with MACE foundation model and simulations of chemical reactor. We combine about 12 000 density functional theory (DFT) calculations with a machine learning workflow that adds over 80 000 additional data points to enable extensive sampling of ground-state and metastable isomers. Our analysis reveals that adsorption energies are primarily determined by local atomic composition of the active site rather than by the precise global-minimum structure. Consequently, a representative set of low-energy isomers provides the same adsorption information as the most stable clusters, which is crucial for experiments where monodisperse samples are difficult to obtain. We further demonstrate that a graphene substrate with a single carbon monovacancy strongly anchors cluster, enhances O₂ adsorption, and lowers CO oxidation barriers by activating traditionally inert Au atoms. Molecular dynamics simulations of CO saturation show that the supported cluster undergoes adsorption-induced reconstruction, with Pt atoms segregating to the surface, and that the support significantly alters the saturation profile compared to the free-standing cluster. This work establishes that rational design of bimetallic catalysts must take into account for local site chemistry, dynamic synergy with the support, and practical relevance of isomer ensembles.

1 | Introduction

The great interest of researchers in metal nanoparticles over the last few decades has been caused by the fact that the transition from micron- to nanosize leads to huge changes in the physical and chemical properties of the materials [1–7]. In this case, the small size of particles is one of the key characteristics, as it leads to an increase in the surface-to-volume ratio and the emergence

of quantum effects that are absent in conventional materials [8]. These effects make metallic nanoparticles very promising for catalysis applications. Discovered by Haruta [9] in 1987, the catalytic activity of gold nanoparticles in the oxidation of carbon monoxide at low temperatures broke the traditional view that gold had no catalytic activity. This attracted a great deal of interest among scientists in studying nanocatalysts, and Haruta found that the small size of the nanoparticles (less than 5 nm), of

nanoparticles was important for the efficiency of heterogeneous catalysis on oxide substrates.

Subsequently, progress in the synthesis of noble-metal nanoparticles made it possible to actively investigate their catalytic activity. In particular, Pt nanoparticles have been shown to be effective in heterogeneous catalysis [10–13]. The activity of single-atom platinum centers in the CO oxidation reaction was also demonstrated, which marked the beginning of a new concept of single-atom catalysts [14]. However, the scarce world reserves of platinum make it much more expensive than other noble metals. Pt-based bimetallic catalysts have been proposed as a promising alternative to very expensive pure Pt-based catalysts. In particular, AuPt NPs have been the subject of intensive study due to their potential as selective oxidation and dehydrogenation [15–17] catalysts and electrocatalysts [18–20]. In particular, the catalytic activity of Au-Pt bimetallic nanoparticles in low-temperature CO oxidation has been shown to be greater than that of the corresponding monometallic catalysts [21, 22]. Several explanations have been proposed for this higher catalytic activity of bimetallic Pt-Au nanoparticles compared to the corresponding monometallic catalysts. Auten et al. [21] suggested that the higher catalytic activity of bimetallic Pt-Au catalysts probably arises from the synergy of Pt and Au, i.e., the catalytically active center probably involves a Pt-Au bimetallic ensemble. On the other hand, Ortiz-Soto et al. [23] argue that the substrate plays a crucial role in improving the catalytic activity of Pt-Au catalysts with respect to the low-temperature oxidation of CO. Wang et al. [22] using DFT calculations showed that Pt sites in bimetallic Pt-Au clusters are catalytically active centres, whereas Au centres act as ‘spectators’ in CO oxidation.

During the last two decades, a significant number of experimental and theoretical studies have been carried out on the geometrical, electronic, magnetic, and catalytic properties of binary Au-Pt cluster [24–36]. Bimetallic Au-Pt small clusters have also been investigated in detail theoretically. In particular, using evolutionary search methods stable structures of alloy clusters with no more than 14 atoms have been found [24, 37, 38]. However, these studies were always limited to the search for stable structures and study of their properties. We conducted a multi-step study to search for all stable structures of AuPt nanoparticles up to 24 atoms using USPEX, calculating adsorption of O and CO on all nonequivalent sites of found particles. In addition, we calculate the CO oxidation barriers on promising nanoparticles located on graphene and on graphene substrate with monovacancy. In this work, we extend the understanding of low-temperature CO oxidation by demonstrating how defect engineering in graphene supports the activation of traditionally inert Au atoms in Pt-Au nanoclusters, leading to enhanced catalytic performance at low temperatures.

2 | Computational Methodology

2.1 | Global Optimization of the Clusters

The global structural optimization of considered PtAu nanoclusters is conducted using the recently developed variable composition mode [39] extending the USPEX code [40, 41]. This method demonstrated a 5–50-fold increase in speed compared

to the traditional USPEX for each composition [39, 42–44]. This approach allows simultaneous finding of optimal structures of nanoclusters within a given range of compositions, effectively harnessing the exchange of structural information between particles of different sizes. As such an exchange is the most efficient when the sizes of particles are close in size, we split the target composition space considered in this study into 4×4 blocks: $\text{Pt}_{1-4}\text{Au}_{1-4}$, $\text{Pt}_{1-4}\text{Au}_{5-8}$, etc. with general formula of $\text{Pt}_{(4p+x)}\text{Au}_{(4q+y)}$, where $p + q \leq 4$, $1 \leq x, y \leq 4$, so that an optimized block is determined by p and q . Thus, the total number of optimized blocks is 15. Optimization generally follows the standard USPEX routine, starting with (1) the creation of 250 initial trial structures using a random symmetry-based method. Next, (2) the structures are locally optimized at the DFT level, and the final energy is used to evaluate their fitness (3). Fitness in case of variable composition is based on the second-order difference in energy over composition [39], which is a measure of local stability (see below). According to the fitness, 70% of the best structures are used as parents for producing (4) the next generation of 200 new structures. These structures are partially derived from previous generations through crossover and mutation operators, and partially consisted of randomly generated candidates to balance exploration and exploitation. We note that it is crossover that allows mixing structural motifs from particles with different compositions. Next, the resulting sample undergoes steps (2) - (4) and the procedure repeats until a stopping criterion is reached. In this study this criterion is stability of the solution over 10 successive generations.

2.2 | Details of DFT Calculations

The total energy calculations and structure relaxations were performed using the VASP code [45, 46]. Our calculations are spin-polarized and use the projector augmented wave method (PAW) [47] and the Perdew–Burke–Ernzerhof (PBE) [48] generalized gradient approximation exchange correlation functional. In these calculations, we use a plane wave kinetic energy cutoff of 450 eV and a supercell method to model finite systems, where periodic images of nanoclusters are separated by a vacuum layer of 12 Å so that the interaction between periodic images vanishes. The orbital occupancies are smeared using the first-order Methfessel-Paxton method with the smearing width of 0.05 eV. The local geometry optimization of considered clusters is conducted until the maximum force acting on each atom is less than 0.03 eV/Å.

Spin-polarized generalized gradient approximation (GGA) with the PBE parametrization is used to calculate adsorption energies. The Grimme correction (DFT-D3) [49] is applied to take into account the dispersive van der Waals interaction. The results of the calculations are post-processed and visualized using the open visualization tool (OVITO) [50] and VESTA package [51, 52]. Electron transfer is explored by the Bader analysis [53].

2.3 | Training of Machine Learning Interatomic Potential for Simulations of Saturation

To study the behavior of Pt-Au clusters in a CO atmosphere, we performed large-scale molecular dynamics simulations using

the moment tensor interatomic potential (MTP) [54]. The MTP decomposes the total energy into contributions from individual atomic neighborhoods and can account for many-body interactions. The atomic energy of each atom depends on the relative position and type of neighboring atoms within a cutoff distance R_{cut} . In our model we employed $R_{cut} = 6 \text{ \AA}$ and level 20 of the MTP potential, readers may refer to Ref. [55] for details. Training of the MTP is performed as follows. First, short DFT molecular dynamics simulations (2 ns) are performed using VASP with PtAu clusters presented in Figure 2 in a constant volume box of dimensions $20 \times 20 \times 20 \text{ \AA}$ with randomly placed 6 CO molecules. MD calculations are carried out at 500 K with a Nose-Hoover thermostat, which allows us to select 522 initial training configurations for MTP.

Subsequently, active learning of MTP using MD simulations is employed to improve the quality of MTP through careful selection of extrapolated atomic configurations, preventing overfitting [55]. Active learning is performed using the mlip package [55] together with LAMMPS [56] for MD simulations to heat Pt-Au clusters from 300 to 673 K and cool from 500 to 10 K in a CO atmosphere. A timestep of 1 fs and a Nose-Hoover thermostat with a damping constant of 10 fs are used for the MD calculations. The simulation box size is the same as that for the initial *ab initio* MD calculations, but we progressively increase the number of CO molecules up to 30. This expands the training set to 1734 different atomic configurations. The performance of the MTP on the training set is shown in Figure S15a. It follows that the mean absolute error (MAE) of energy prediction is 8.7 meV/atom. To assess the performance of MTP over atomic configurations not included in the training set, we construct a validation set by performing AIMD calculations on different PtAu clusters containing 6 CO molecules in the $20 \times 20 \times 20 \text{ \AA}$ box. This set comprised 8264 atomic configurations. The corresponding MTP performance is shown in Figure S15b. The MAE = 7.1 meV/atom for the validation set is in good agreement with the value obtained for the training set. There are different clustered points corresponding to different PtAu configurations, which can be contrasted with amore uniform distribution of points in the training set in Figure S15a obtained during active learning procedure.

2.4 | Machine-Learning Workflow for Screening of Adsorption Energy

To accelerate the evaluation of adsorbate configurations on PtAu clusters, we developed a two-stage machine-learning workflow combining a transferable MACE foundation potential with a task-specific energy model. The objective of this approach is to reduce the number of required density functional theory calculations by enabling rapid geometry relaxation and energy ranking of candidate adsorbate-cluster configurations.

The training dataset is constructed from DFT-relaxed PtAu clusters containing either CO or atomic O adsorbates. For each configuration, the relaxed geometry is taken from the VASP [45, 46] calculations, and the corresponding DFT total energy is extracted from the final electronic step. This produced a dataset of 12154 adsorbate-cluster configurations spanning CO

and O adsorption on PtAu clusters. To avoid information leakage between structurally similar configurations, the dataset is split by cluster identity, rather than individual adsorption configurations. As a result, the train, validation, and test sets consisted of 9417, 1689, and 1977 structures, respectively.

The energy model is obtained by fine-tuning a pretrained MACE foundation model [57] on the DFT total energies of relaxed structures. The model is trained in an energy-only regime, using the DFT energy as the sole supervised target. Force and stress contributions are excluded from the loss function. Training is performed with an energy weight of 1.0, a learning rate of 5×10^{-4} , a batch size of 32, exponential moving average regularization, and early stopping with a patience of five epochs. Isolated atomic energy offsets are estimated using the average-energy option implemented in MACE.

The resulting model achieved a test-set energy RMSE of 13.9 meV/atom, as reported by the MACE evaluation routine. An independent post-training evaluation on the held-out test set gave a total-energy MAE of 0.185 eV. When normalized by the number of atoms in each cluster-adsorbate system, the mean absolute error is 9.98 meV/atom. These results indicate that the fine-tuned model accurately captures the relative DFT energy landscape of PtAu-adsorbate systems.

For inference on previously unseen structures, we used a two-step procedure. First, the initial unrelaxed geometries are relaxed using the publicly available pre-trained MACE-MP foundation potential [57], which provides force predictions suitable for geometry optimization. Cluster calculations are performed with periodic boundary conditions disabled, and each structure is placed in a cubic simulation box of $30 \times 30 \times 30 \text{ \AA}$. Geometry optimization is carried out using the FIRE optimizer until a force threshold of 0.05 eV/ $\text{\AA}$ is reached, or up to a maximum of 500 optimization steps. Second, the energy of the MACE-relaxed geometry is evaluated using the fine-tuned energy model described above. This separation between geometry optimization and energy prediction is chosen because the fine-tuned model is trained only on energies and is therefore not used as the force model for relaxation. Instead, the pretrained MACE-MP model provided a transferable relaxation engine, while the fine-tuned model supplied a system-specific energy estimate calibrated to the DFT data for PtAu clusters with CO and O adsorbates. In this way, the workflow combines the robustness of a broadly trained foundation potential with the accuracy of an energy predictor adapted to the task.

2.5 | UMAP Visualization

In order to discover the configurational space of active sites of CO and O on the PtAu nanoclusters the uniform manifold approximation and projection for dimension reduction (UMAP) model [58] is used. For the projections we use the “small” MACE model [59] to calculate descriptors without dispersion correction. This model is represented by higher order equivariant neural network, that takes into account equivariant geometrical representation of sites. Message passing procedure allow us

consider neighbors of one atom and aggregate information about some behavior of atomic site. This allows for a sufficient depiction using descriptors obtained as values of the neural network output layer, which can then be subjected to a compression procedure. We use weight from model, pretrained on structures, containing up to 89 elements and calculated with DFT (PBE+U) level of theory. These weights can be loaded from <https://github.com/acesuit/mace> repository. After that, UMAP model is trained using these MACE descriptors. For determining the site composition, we use an algorithm, considering neighbors of atoms in adsorbate molecules (O and CO respectively). We identify neighbors on the surface to which the corresponding adsorbate molecules are approached (the cutting radius is 5 angstroms). Also we calculate the convex hull of the cartesian position of particles to verify that all adsorbate atoms are outside. After calculations of structural descriptors the UMAP model with a number of neighbors of 1000 and a minimal distance of 0.2 is used for data clustering. These UMAP hyperparameters are sufficient for a good visual separation of the data.

3 | Results and Discussion

3.1 | USPEX Prediction

The ground-state structures of Au_nPt_m clusters are obtained using the USPEX evolutionary algorithm. The most stable structures among all calculated Au_nPt_m clusters ($n + m \leq 24$) are shown in Figure 1. Previously, several studies have investigated small AuPt clusters using various approaches, including substitution of one or two Pt atoms in Au clusters [27, 30, 60], construction of variable-composition structures based on known Au or Pt cluster motifs [24–26, 29], systematic composition variation for fixed cluster size (Au_xPt_y , $x + y = 10$) [38], and evolutionary algorithms such as the IM2ODE package [37].

As the cluster size increases, the structural landscape becomes increasingly complex, and discrepancies between different theoretical studies become more pronounced. To provide a rigorous quantitative assessment, we re-relaxed all literature-reported structures using our identical DFT setup and compared them with our USPEX-generated isomers. The comparison for six representative compositions (Au_5Pt_3 , Au_6Pt_3 , Au_8Pt_2 , Au_4Pt_5 , Au_2Pt_4 , and AuPt_9) is presented in Figure S1, which shows the eight lowest-energy isomers for each composition together with the re-relaxed literature structures and their relative energies ΔE (in eV).

Our comparison shows that the stable structures reported in previous works can be found among our low-energy isomers with relative energies up to about 0.7 eV above the global minimum. Such moderate energy differences are not surprising and can be attributed to variations in computational parameters such as exchange-correlation functionals, pseudopotentials, and convergence criteria used in different studies. Detailed energetic rankings for each composition are provided in Figure S1.

Several compositions exhibit remarkably flat potential energy surfaces with multiple nearly degenerate isomers. For example, in Au_4Pt_5 , five structurally distinct isomers lie within an energy

window of only 1 meV, despite having pairwise RMS fingerprint distances of 0.02–0.15 Å. This near-degeneracy highlights the difficulty in identifying the true global minimum and underscores the importance of considering ensembles of isomers for practical applications. The structural trends across these compositions are noteworthy: Au_2Pt_4 remains planar (2D) for all nine lowest-energy isomers, while AuPt_9 is exclusively non-planar (3D). Au_8Pt_2 shows a clear separation into a compact 3D low-energy family and a planar 2D higher-energy family, whereas Au_6Pt_3 exhibits the unusual behavior of having non-planar low-energy isomers and planar higher-energy ones.

Analyzing the cluster structures, we observe that Pt atoms in ground-state AuPt clusters tend to occupy the most highly coordinated positions, predominantly located at the cluster core, while Au atoms reside at the edges or vertices. This segregation is rationalized by the relative bond strengths: Pt–Pt (1.58 eV) > Pt–Au (1.27 eV) > Au–Au (1.15 eV). Concentrating Pt atoms at the centre maximizes Pt–Pt and Pt–Au bonds, yielding more compact and stable structures. This arrangement facilitates *d*-electron mobility between Pt atoms, creating small aromatic regions with free *d*-electrons in the Pt blocks. In contrast, the localized nature of Au’s *d*-electrons restricts their involvement in multi-centre bonding, favoring placement of Pt in the core, where they can effectively form multi-center bonds with neighboring Pt atoms [37, 38]. Owing to the Jahn–Teller distortion, most clusters slightly distort toward lower symmetry [37].

To investigate stability across the whole compositional range, we employed the criteria of local or conditional stability. We calculate the second-order finite differences in the total energy with respect to the number of atoms and identified their minimum:

$$\Delta_{\min}^2(n, m) = \min\{\Delta_{\text{Au}}^2(n, m), \Delta_{\text{Pt}}^2(n, m)\} \quad (1)$$

where

$$\Delta_{\text{Au}}^2(n, m) = E(n + 1, m) + E(n - 1, m) - 2E(n, m) \quad (2)$$

$$\Delta_{\text{Pt}}^2(n, m) = E(n, m + 1) + E(n, m - 1) - 2E(n, m) \quad (3)$$

and $E(n, m)$ is the ground-state energy of the Au_nPt_m cluster. The value of $\Delta_{\min}^2(n, m)$ serves as a measure of stability; clusters with positive values are termed “magic”. Numerically, $\Delta_{\min}^2(n, m)$ characterizes the stability against the mixing of one atom of each type between two identical Au_nPt_m clusters. This criterion has been previously applied to binary systems, as discussed in Ref. [61–63]. In experimental synthesis, clusters are typically produced as ensembles, and the magnitude of $\Delta_{\min}^2(n, m)$ can serve as an indicator of cluster abundance. Stable clusters form in a certain sequence, most likely related to the tendency towards closed-shell electronic structures, as demonstrated for CuAu clusters by Mikhailova et al. [64]. Analysis of the twelve most stable clusters, presented in Figure 2, reveals a consistent structural motif: Au atoms systematically occupy peripheral and surface sites, forming an outer shell around a Pt-rich core. This segregation is driven by the lower surface energy of Au compared to Pt, a well-documented phenomenon in bimetallic systems that promotes the formation of stable core–shell architectures.

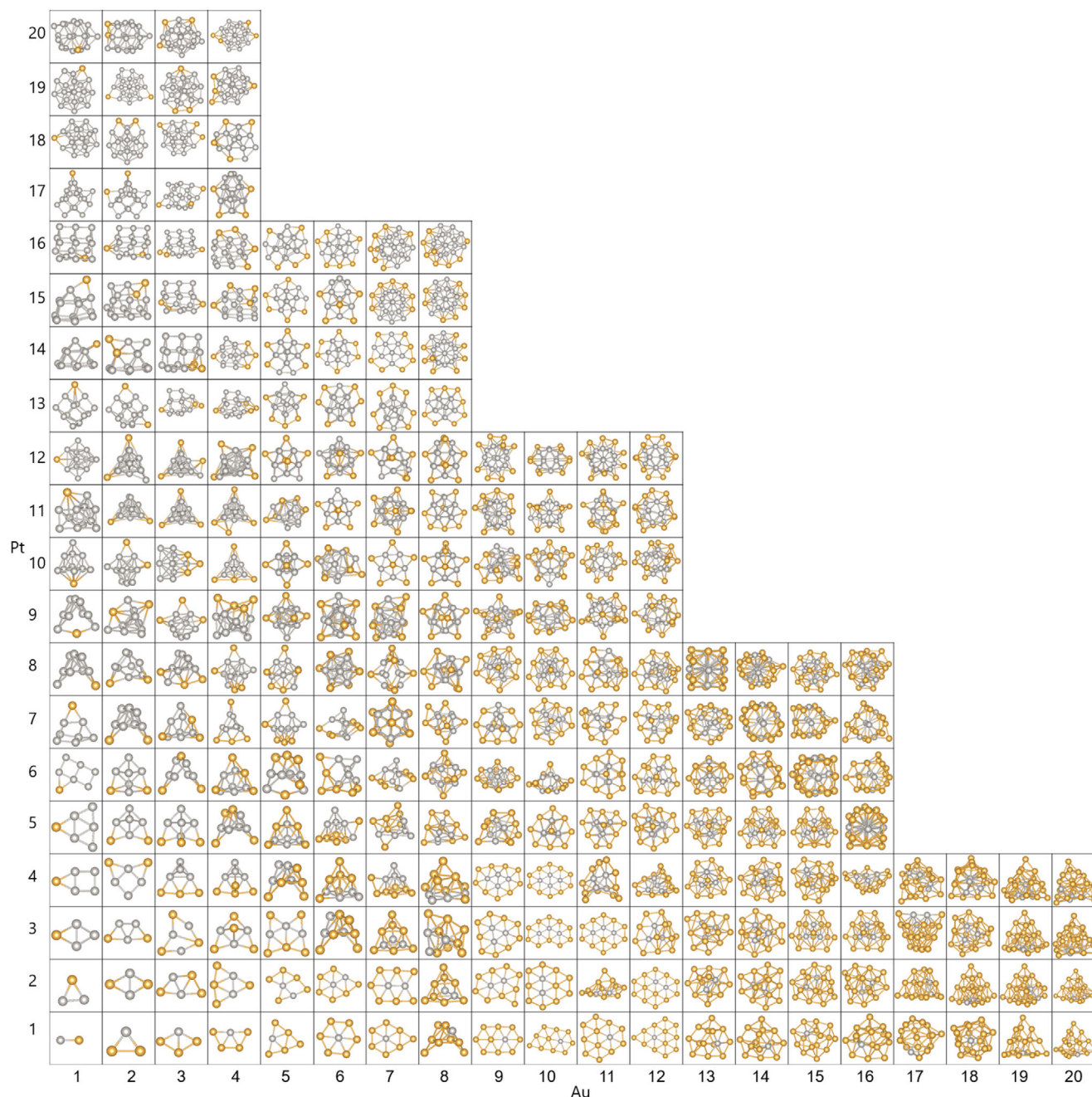


FIGURE 1 | The most stable structures of the Au_lPt_m ($l + m \leq 24$) cluster (240 clusters), where Au and Pt atoms are represented by yellow and grey balls, respectively.

3.2 | Adsorption Properties

To efficiently screen for catalytic activity without performing computationally intensive reaction path calculations, researchers often employ structural descriptors. A key example is the coordination number (CN) of surface atoms, which serves as a proxy for reactivity. This concept has been extensively validated in the literature. Ma and Xin introduced the orbital-wise coordination number as a robust descriptor for predicting adsorption properties of metal nanocatalysts [65], while Calle-Vallejo and co-workers developed the generalized coordination number ($\overline{\text{CN}}$) for fast prediction of adsorption properties on platinum nanocatalysts [66], demonstrated its application in identifying optimal

surface sites for heterogeneous catalysis [67], and recently systematized the use of generalized coordination numbers as a descriptor in electrocatalysis [68]. Atoms with low coordination numbers are typically under-coordinated and possess enhanced catalytic activity, as their unsaturated bonds facilitate stronger interactions with adsorbates. This approach is grounded in the principles of the Sabatier principle and Brønsted–Evans–Polanyi (BEP) relationships, which link surface geometry to adsorption strength and reaction barriers, allowing for rapid identification of promising catalyst candidates. However, this method performs poorly in bimetallic clusters as it does not consider the type of atom. The d -band model developed by Hammer and Nørskov [69] correlates transition metal reactivity with the energy center of

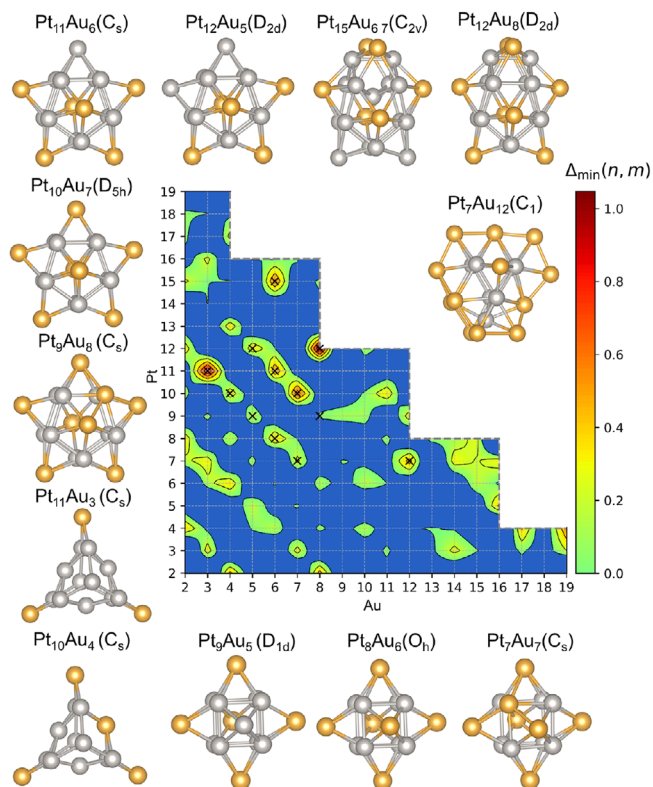


FIGURE 2 | Heat map of $\Delta_{\min}^2(n, m)$ and atomic structure of twelve stable PtAu clusters. Pt atoms are shown by grey color, while Au atoms are orange.

d-states of surface atoms. As previously reported, this approach accurately quantifies variations in adsorption energy for atomic and molecular species at different nanoparticle surface sites [66]. However, as shown in our recent paper [70], this descriptor also correlates poorly with adsorption energy in complex disordered structures.

To evaluate the catalytic efficiency of the PtAu clusters identified through our USPEX calculations, we calculate the adsorption energies of O and CO on each non equivalent surface active site. This high throughput screening required extensive DFT calculations. Small clusters have only a few adsorption sites, but a 24 atom cluster has approximately 40 different sites. In total, characterizing adsorption properties across a set of globally stable nanoparticles requires 6157 DFT calculations for atomic oxygen and 5997 DFT for CO, resulting in about 12154 combined DFT calculations overall.

To extend this analysis beyond the most stable stoichiometries, we also consider nine higher energy isomers for each cluster size, ranked by increasing total energy relative to the global minimum (see Figure S2). For these isomers, we employ a two stage machine learning workflow to efficiently predict adsorption energies (more details in the Supporting Information). First, a pre-trained MACE foundation potential performs geometry relaxation of adsorbate cluster configurations. Second, a fine tuned energy model trained on DFT total energies of 12 154 relaxed Pt-Au adsorbate structures provides accurate single point energy predictions. This approach yields an additional 40 272 predicted adsorption energy values for oxygen and 40 034 for

carbon monoxide, dramatically expanding our screening capabilities. Importantly, the adsorption energy distributions for the nine isomers closely resemble those of the corresponding most stable clusters. This indicates that local adsorption properties are primarily determined by the local atomic environment of the active site rather than by the precise global minimum structure.

The resulting adsorption energy distributions from both DFT and ML are visualized as heat maps in Figure 3. Panels a, b and c show the minimum, average, and maximum adsorption energies for O from DFT. Panels d, e and f show the same for ML. Panels g, h and i present corresponding data for CO from DFT, while Panels j, k and l present the results from ML. The crosses in the DFT panels indicate the 12 most stable clusters from Figure 2. Our analysis reveals that the adsorption energies of O and CO on stable PtAu clusters lie within a moderate range, neither being excessively strong nor weak. This finding is significant because the Sabatier principle states that optimal catalysts should bind reactants neither too strongly nor too weakly. The observed adsorption properties, further supported by consistent ML predictions for both stable and metastable clusters, strongly suggest that these small PtAu nanoparticles have great potential as catalysts for CO oxidation.

Comparing the DFT and ML results for oxygen adsorption, the heat maps of minimum and average energies show excellent agreement between the two methods. However, the maximum adsorption energies are notably higher for ML than for DFT, and this small deviation is confined to the upper tail of the distribution. To understand its origin, we decompose all adsorption energies according to the composition of the adsorption site. The ML model yields slightly more positive adsorption energies for sites consisting of a single gold atom. This discrepancy arises from the greater structural diversity present in the isomer set compared to the global minimum clusters. These isomers contain single gold atom sites with a wider range of local coordination environments, including more highly coordinated or sterically hindered positions. The ML model, trained on this diverse pool of configurations, captures the subtle geometric and electronic effects that weaken adsorption on isolated gold atoms. In contrast, the global minimum clusters predominantly feature gold atoms at low coordination edges or vertex sites, where DFT intrinsically predicts stronger binding. The excellent agreement for the average and minimum values confirms that the overall adsorption trends are captured correctly, while the slight difference in maximum values reflects a more complete sampling of weakly binding Au sites in the ML dataset. Figure 4 presents a detailed breakdown of oxygen adsorption energies from both DFT and ML as a function of site composition and type, clearly illustrating this effect.

As systematically plotted in Figure 4, the adsorption energy for oxygen exhibits a pronounced and systematic dependence on the local composition and coordination environment of the adsorption site. A clear trend emerges, with sites primarily composed of platinum atoms yielding strongly favorable (more negative) adsorption energies, indicating robust chemisorption. In stark contrast, sites with high gold content show significantly weaker binding, as evidenced by markedly less negative adsorption energies. This fundamental difference is attributed to the relatively inert nature of gold compared to the high reactivity of

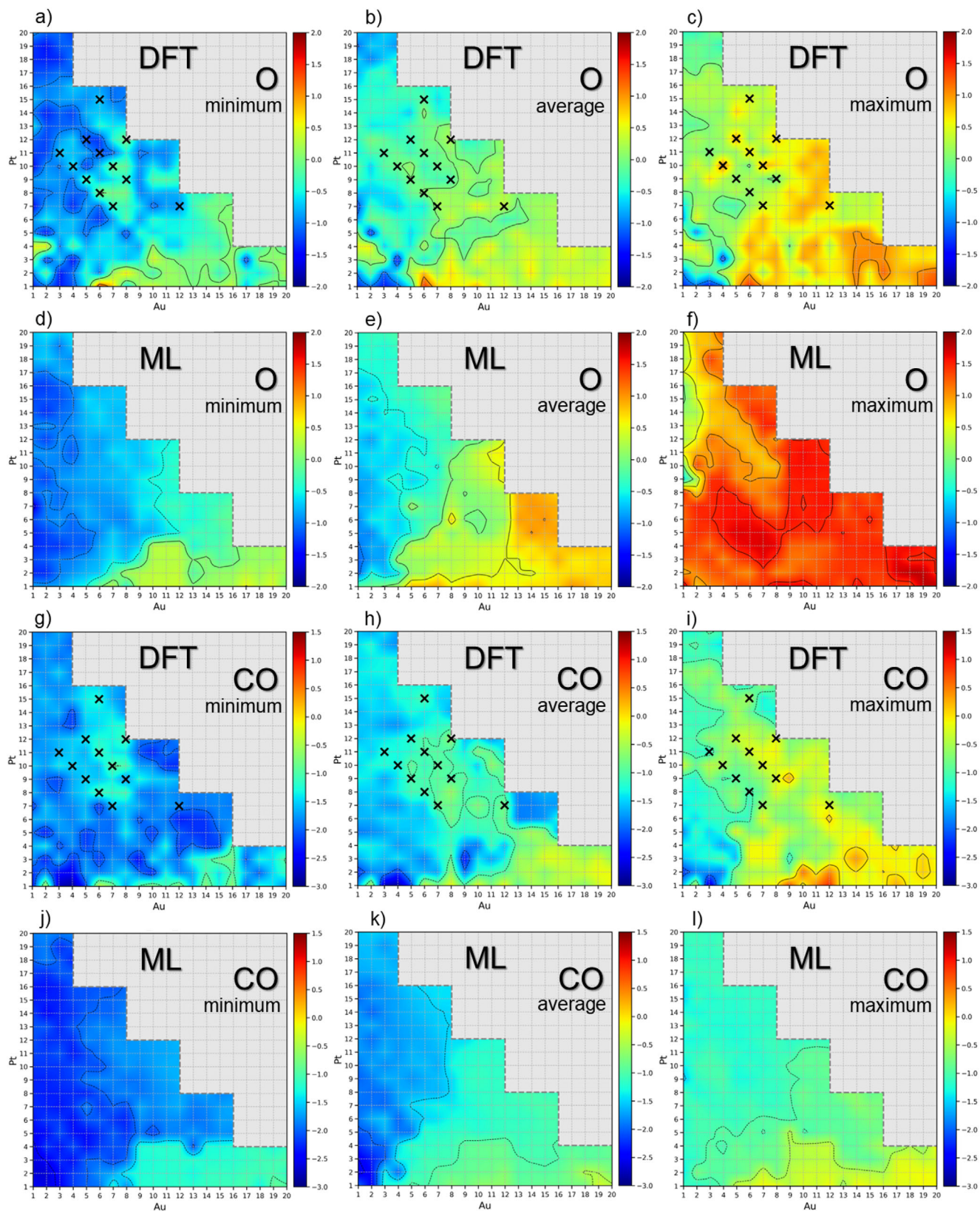


FIGURE 3 | Heat maps of minimum, average and maximum adsorption energies of O and CO on PtAu clusters from DFT and ML.

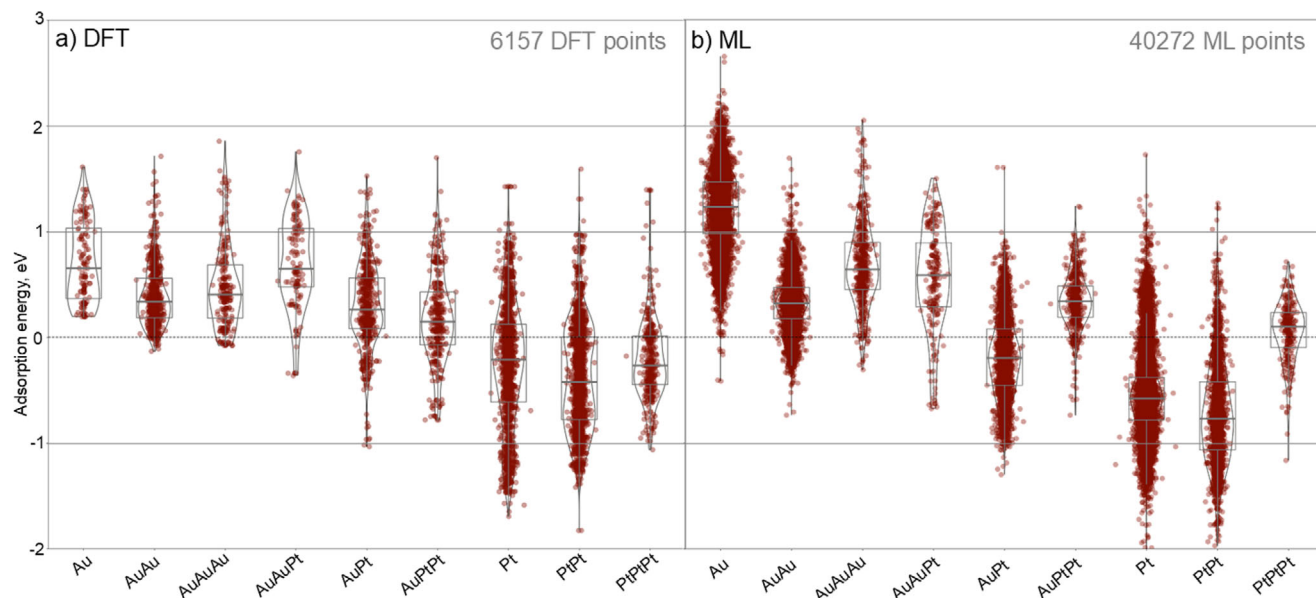


FIGURE 4 | Adsorption energies of O atom on the considered bimetallic PtAu clusters of different compositions and types of adsorption sites from (a) DFT and (b) ML calculations.

platinum. These findings on PtAu systems are directly consistent with our previous works [71–73] on PdAu and CuAu bimetallic nanoparticles, where a similar principle was observed. The adsorption strength of simple diatomic and atomic adsorbates is predominantly governed by the local electronic structure of the specific surface site, which itself is dictated by the identity of its constituent atoms.

The obtained results can be clearly analyzed using uniform manifold approximation and UMAP visualization, as shown in Figure 5. UMAP is plotted for all considered AuPt nanoclusters (including ground states and isomers), with adsorbed oxygen atoms (top panel of Figure 5) and CO molecules (bottom panel). The projections of DFT and ML-based data on MACE structural descriptors are shown in Figure 5a,b. Comparison of these data distributions clearly demonstrates that additional data on PtAu isomers expands the regions covered by DFT data. All active adsorption sites considered for ground-state clusters are located within regions corresponding to isomers. In general, it means that it is not necessary to consider only ground-state clusters to study adsorption properties of small nanoclusters. It is sufficient to consider a set of metastable isomers which are likely to occupy the same region in configuration space. This will give the same information about the adsorption properties of nanoclusters in the target system. Considering the UMAP visualization of adsorption energies for each active site in clusters (Figure 5c,d), one can see that these data agree well with the data from Figure 4, where the distribution of DFT data for adsorption energies is located within the wider distribution of other metastable clusters.

It is also important to consider the size of nanoparticles. Visualization of these nanoparticles is presented in Figure 5e,f for O and CO adsorbates, respectively. Comparison of the sizes of the clusters with adsorption energy distributions indicates that all clusters may have high or low adsorption energies. The most significant factor is the composition of each adsorption site, as can be seen by comparing Figure 5g,h with Figure 5c,d. Sites with

the highest adsorption energies for O correspond to Au-rich sites (red, orange and yellow colors in Figure 5g), while sites with the lowest adsorption energies correspond to Pt-rich active sites (blue regions in Figure 5g). The same is true for CO adsorption, where Au-rich sites have higher adsorption energies than Pt-rich ones (Figure 5h).

Due to the considerable computational cost associated with screening a large number of adsorption sites, the positions of atoms in PtAu clusters are initially fixed during geometry optimization; only the degrees of freedom of adsorbates are relaxed. To quantify the error introduced by this constraint, a systematic comparison is performed for adsorption of O and CO on 12 stable clusters. Full structural relaxations are conducted for these cases where both the adsorbate and atoms in the cluster are allowed to move freely. The results are summarized in Figure S3, which presents adsorption energy values as Gaussian distributions for both the fixed-cluster and full-relaxation calculations. The data reveals a systematic and significant deviation: the adsorption energies derived from fully optimized structures are consistently more negative, indicating stronger binding. There is a uniform shift in adsorption energies of approximately 0.3 to 1 eV for O and from 0.3 to 0.8 eV for CO across the dataset (see Figures S4–S6). This demonstrates that the fixed-cluster approach systematically overestimates adsorption energy, makes it less negative. The finding are in excellent agreement with the work of Ma et al. [65], who established that constraining nanoparticle atoms during optimization suppresses the substrate’s ability to relax and stabilize the adsorbate, leading to overestimated (less favorable) adsorption energies. Our results confirm that while the fixed-atom approach is a necessary approximation for high-throughput screening, it introduces a predictable and consistent error that must be considered when interpreting absolute adsorption strengths.

The results presented above lead to an important practical conclusion. The adsorption energies for O and CO are primarily

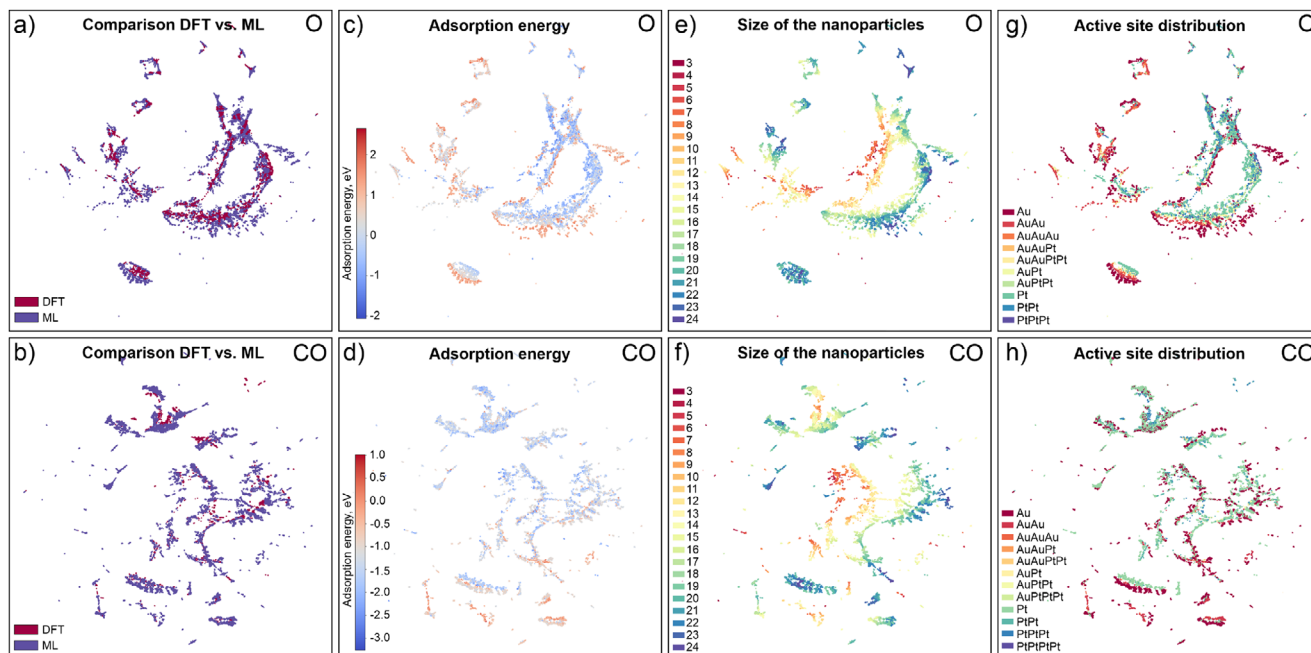


FIGURE 5 | (a,b) Uniform manifold approximation and projection (UMAP) visualization of the data ($n_{neighbors} = 1000$, $d_{min} = 0.2$) showing the comparison between data obtained by DFT and the MACE model for PtAu nanoparticles (DFT is used for the ground state, and ML is used for isomers). Adsorption energies of (c) O and (d) CO on each active site of each considered PtAu nanocluster. (e,f) The size of the nanoparticles ranges from 3 to 24 atoms, and the composition of each active site is shown for (g) O and (h) CO adsorption. Each point represents an individual adsorption active site. The embeddings reveal the overall similarity in the structure of adsorption sites.

governed by the local atomic composition and coordination environment of the active site, not by the specific global minimum structure of the cluster. This is clearly demonstrated by UMAP analysis, where DFT data for ground state clusters lie entirely within the broader distribution covered by the isomers. Consequently, a full set of metastable isomers provides essentially the same information about adsorption properties as global minima. This finding is particularly valuable for experimental synthesis, as obtaining perfectly monodisperse populations of single most stable clusters is extremely challenging. Instead, realistic samples inevitably contain a distributions of isomers, which are perfectly adequate for studying catalytic adsorption as they explore the same range of local environments and adsorption energies. Therefore, exhaustive global optimization may not be strictly necessary for screening the adsorption properties of small bimetallic clusters. A representative set of low energy isomers can serve as a reliable and computationally more accessible proxy.

3.3 | Graphene Substrate

Since the clusters involved in catalytic reactions are located on substrates, in this work we also considered clusters supported on graphene. Pristine graphene contains various types of intrinsic and extrinsic defects, among which monovacancies are one of the most common and technologically relevant point defects [74]. This work investigates the mobility and stability of a model PtAu bimetallic cluster on both pristine graphene and graphene featuring a carbon monovacancy. Previous studies have demonstrated that monovacancies in graphene serve as powerful anchoring sites for transition metal clusters, significantly enhancing the metal-support interaction through strong

TM-C bonds [75, 76]. Our computational analysis reveals a stark contrast in cluster mobility between the pristine and defective substrates. For pristine graphene, the potential energy surface for cluster migration is remarkably flat, characterized by a very low diffusion barrier of approximately 0.4 eV (see Figure S7b). This minimal energy requirement is easily overcome at elevated temperatures, indicating that PtAu clusters will exhibit high mobility and a strong tendency to coalesce into larger particles, ultimately leading to catalyst degradation.

In contrast, the introduction of a monovacancy dramatically alters the cluster-substrate interaction. As shown in Figure S7e, the cluster becomes trapped in a deep energy well at the defect site. Due to the large depth of this well, thermal processes are not sufficient to facilitate escape at relevant temperatures for catalytic reactions. This trapping phenomenon can be explained by the unsaturated carbon bonds around the monovacancies, which provide favorable sites for metal atoms in the cluster [77, 78]. This strong chemisorption effectively anchors the cluster, preventing migration and sintering. Furthermore, recent studies have shown that electron penetration from graphene supports through vacancy defects can triggers interfacial activity, significantly improving catalytic performance for CO oxidation reactions at room temperature [79].

To assess the influence of substrate interactions, we investigated how a graphene support affects the CO adsorption properties of a Pt_7Au_7 cluster. This composition was selected as a representative model system for detailed mechanistic studies for two reasons: (1) it possesses an equal Pt:Au ratio (1:1), which allows for systematic comparison of the catalytic roles of both metal species, and (2) it exhibits high stability according to our $\Delta^2_{min}(n, m)$

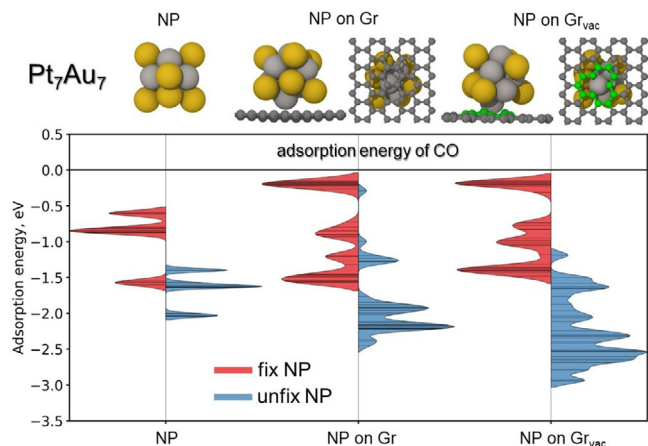


FIGURE 6 | The distributions of adsorption energy of CO on the Pt_7Au_7 cluster in three environments: free-standing (NP), supported on pristine graphene (NP on Gr), and supported on graphene with a vacancy defect (NP on Gr_{vac}) with fixed (red) and unfixed (blue) metal atoms. Atoms around the monovacancy are highlighted by green.

analysis presented in the heat map of Figure 2. Additionally, prior to performing these calculations, we examined several distinct adsorption positions of the cluster on both pristine and defective graphene surfaces and selected the configuration with the lowest total energy, ensuring that the supported cluster geometry is representative of the most favorable cluster-support interaction. We computed adsorption energies at approximately 40 distinct sites for three systems: the free-standing cluster, the cluster on pristine graphene, and the cluster on defective graphene. Calculations were performed with both a fixed cluster geometry and with full structural optimization. As shown in Figure 6, the graphene substrate significantly strengthens CO adsorption (makes the energy more negative). The average adsorption energy was -1.66 eV for the free cluster, which decreased to -1.89 eV on pristine graphene and -2.25 eV on defective graphene. This enhancement is attributed to substrate-induced structural changes in the cluster, as proven by the nearly identical adsorption energies obtained when the cluster's structure was fixed. Furthermore, our calculations of migration barriers indicate that defects act as anchoring sites for clusters. These findings are critical for catalytic applications. They demonstrate that the support is not inert but actively tunes the reactive properties of the cluster.

3.4 | Reaction Barriers of CO Oxidation

The preferential oxidation of carbon monoxide is a critical purification step in the purification process of hydrogen production through fuel reforming. This process drives the ongoing search for highly active and selective catalysts [80]. Here, we study the oxidation reaction of CO using the Langmuir-Hinshelwood (LH) mechanism with graphene-supported Pt-Au nanocluster as a model system. We focus our analysis on a single representative Pt_7Au_7 cluster to carefully explore various reaction pathways. This approach enables a comprehensive investigation of initial configurations of reactants on different active sites of the cluster, including: CO on Pt with O_2 on Pt, CO on Au with O_2 on Pt, CO on Pt with O_2 on Au, and CO on Au with O_2 on Au. The Pt_7Au_7

cluster is studied in two distinct environments: supported on pristine graphene and on graphene with a monovacancy defect.

Our calculations reveal that the defect state caused by the presence of a vacancy in graphene has a decisive influence on the adsorption properties and catalytic activity of the entire Pt_7Au_7 cluster. A key finding is the stark contrast in behavior between nanoparticles anchored to defective graphene and pristine graphene. On pristine graphene, O_2 exhibits strong preferential adsorption on Pt sites, while CO adsorbs on both Au and Pt. However, this site-specificity vanishes when the cluster is anchored to a graphene vacancy. We attribute this altered behavior to a significant electron density redistribution induced by the defective substrate, which effectively modulates the catalytic properties of the active sites. For example the O_2 adsorption energies on Pt and Au atoms within a Pt_7Au_7 cluster are significantly enhanced when the cluster is supported on defective graphene with a monovacancy, as shown in Figure 7a. To elucidate the origin of this enhancement, we analyzed the adsorption on a specific Au atom. Electron density redistribution plots (Figure 7b,c) reveal that the monovacancy induces a strong local electron density redistribution, increasing the electron transfer from the Au atom to the O_2 molecule (from $0.27e^-$ on pristine graphene to $0.4e^-$). Crystal Orbital Hamiltonian Population (COHP) analysis further reveals that this increased charge transfer strengthens the Au-O bond, accounting for the more negative adsorption energy of O_2 (Figure 7d,e). Crucially, the excess electrons populate the O_2 antibonding orbitals, facilitating bond cleavage [77]. Thus, graphene defects promote both stronger O_2 adsorption and a higher probability of O_2 dissociation, which collectively lower the kinetic barriers for CO oxidation.

This profound substrate effect challenges previous assumptions regarding the role of Au in bimetallic systems. For instance, Wang et al. [22], in their study of unsupported ultrasmall $\text{Pt}_m\text{Au}_{4-m}$ clusters, concluded that Au atoms primarily serve a structural role in preventing CO poisoning without actively participating in the catalysis. In contrast, our results demonstrate that when supported on a defective graphene substrate, the electronic structure of the entire cluster including the Au atoms is modulated, suggesting a more synergistic and active role for Au than previously recognized.

This electronic modulation directly impacts the catalytic activity, as evidenced by the calculated energy barriers for the CO oxidation reaction. If Pt atoms are inherently more active, one would expect significantly lower barriers when both O_2 and CO are adsorbed on Pt sites compared to scenarios involving Au sites. However, as shown in Figure 8, the energy barriers for reactions where both reactants are on Pt sites are comparable to, or even higher than, those where one molecule is adsorbed on an Au atom (see red curve in Figure 8a,b). This indicates that the electron donation from the defective graphene enhances the reactivity of the Au atoms, making them significant participants in the catalytic process. Furthermore, a systematic analysis of all reaction pathways (see Supporting Information, Tables S1, S2 and Figures S8-S13) reveals a consistent trend: the energy barriers for clusters on defective graphene are generally lower than those on pristine graphene (Figure 9). This can be explained by the additional electron density donated by the graphene vacancy, which facilitates reactions at various active sites.

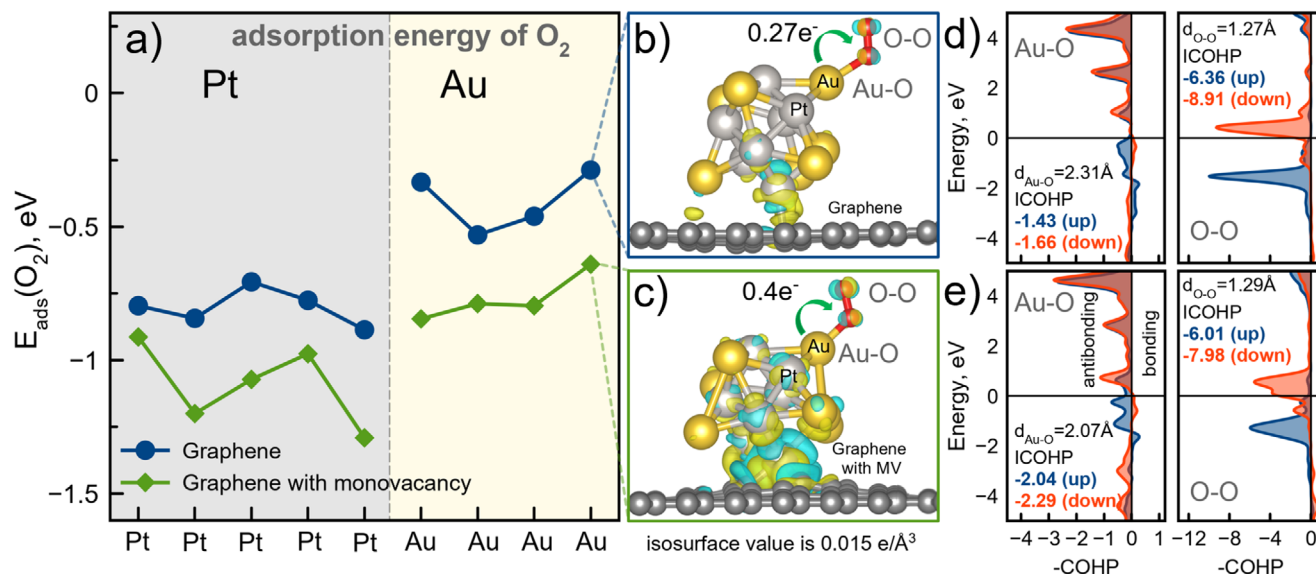


FIGURE 7 | (a) The adsorption energy of O₂ on Pt₇Au₇ cluster on graphene (blue) and graphene with vacancy (green) computed at different adsorption sites of Pt₇Au₇ cluster, namely Pt or Au. (b,c) The atomic structure of the Pt₇Au₇ cluster with adsorbed O₂ with charge redistribution explicitly shown. (d,e) Calculated COHP diagrams projected on different spins for these two cases show a significant difference in chemical bonding when defective graphene is used as a support.

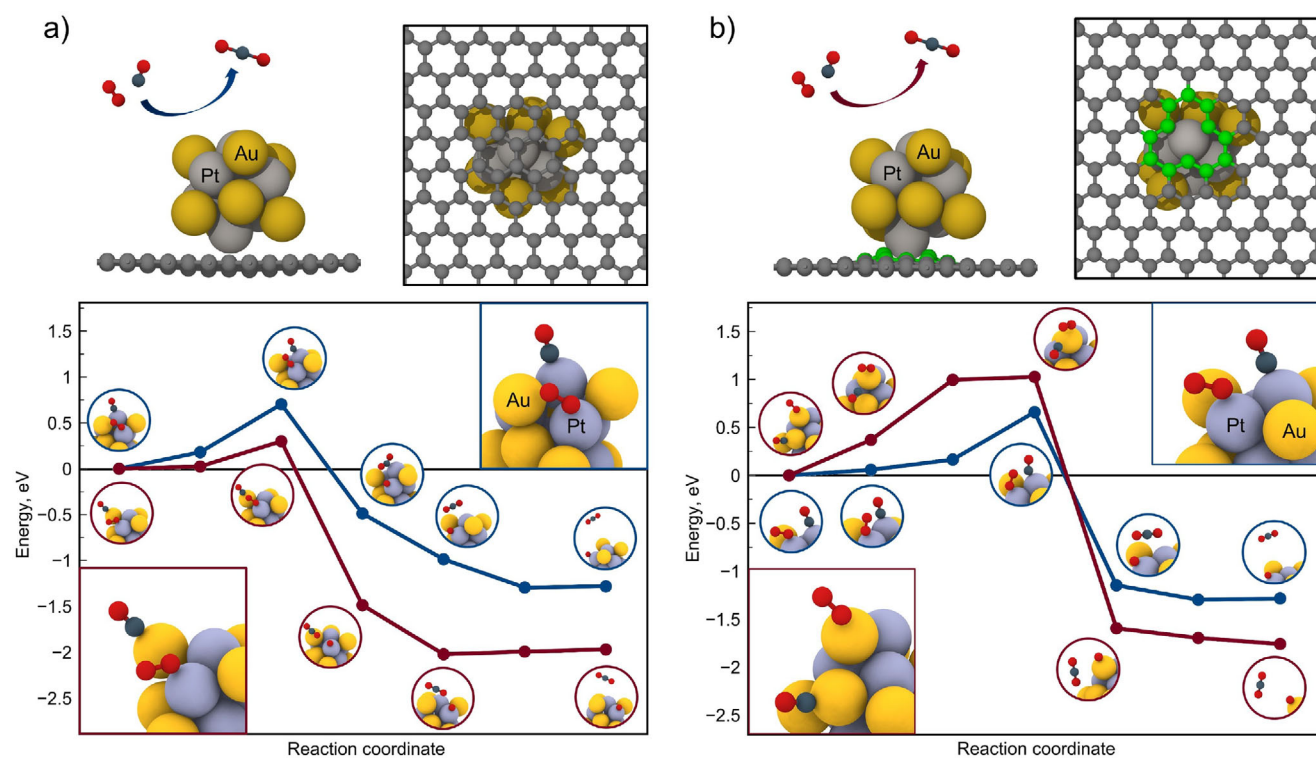


FIGURE 8 | The energy barriers for the CO oxidation reaction on Pt₇Au₇ cluster placed on (a) defectless graphene and (b) graphene with a vacancy. Energy barriers are calculated for two different adsorption sites for the Pt₇Au₇ cluster. The barrier for the Pt adsorption site of CO is shown in blue, while the Au adsorption site of CO is in red.

In addition to the typical reaction profiles shown in Figure 8, the full set of calculated pathways reveals a broader change in the kinetic landscape caused by the graphene vacancy. For Pt₇Au₇ supported on pristine graphene, CO oxidation barriers vary in the range of 0.19–3.30 eV with an average of 1.34 eV. In

contrast, for Pt₇Au₇ anchored on a graphene monovacancy, the barriers shift to lower values, varying in the range of 0.13–3.02 eV with an average value of 1.03 eV. The number of pathways with barriers below 1 eV increases from 12 for pristine graphene to 18 for defective graphene (Figure 9). Thus, the role of the

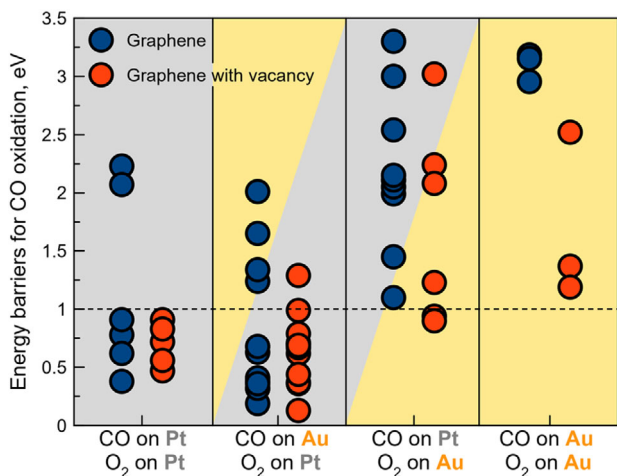


FIGURE 9 | The energy barriers of CO oxidation on Pt₇Au₇ cluster on graphene (blue points) and graphene with vacancy (red points) computed for different active sites of the nanocluster. Four sets of data correspond to different adsorption sites for CO and O₂ during the calculations of the CO oxidation reaction, namely CO on Pt, O₂ on Pt (gray region), CO on Au, O₂ on Pt (gold and gray region), CO on Pt, O₂ on Au (gray and gold region), and CO on Au, O₂ on Au (gold region).

monovacancy is not limited to stabilizing cluster on the surface, but rather it changes the overall distribution of reaction barriers and increases the number of kinetically accessible pathways for CO oxidation. It should be noted that the lowest-barrier pathways are not necessarily related to traditional reactions involving only Pt. On pure graphene, the minimum barrier of 0.19 eV is achieved when CO is adsorbed on an Au atom and O₂ on a Pt atom. A similar mixed-metal configuration yields the lowest barrier of 0.13 eV for a cluster with vacancies. This result indicates that Au atoms can directly participate in kinetically favorable reaction pathways when their electronic structure is modulated by the environment of the Pt-Au cluster and, to a greater extent, by the defective graphene substrate. Thus, the vacancy promotes a more distributed catalytic response, in which Au-containing interfacial regions also contribute to CO oxidation. At the same time, the calculated barriers show that not all Au-related pathways are equally favorable. Configurations where O₂ is adsorbed on Au sites still include several high-barrier channels, especially when both reactants are bound to Au atoms. Therefore, the key effect of vacancy in graphene should be viewed as a redistribution and expansion of low-barrier pathways, rather than uniform activation of all Au sites. This distinction is important because it links enhancement of catalytic activity to specific local metal-substrate configurations rather than to average Au content in the nanoclusters.

Thus, our observations highlight the dual importance of nanocluster size and supporting material in catalysis. The substrate is not merely an inert scaffold, but an active component that can dramatically alter adsorption behavior and catalytic properties, even activating traditionally less-active metal atoms such as Au. This insight emphasizes that the rational design of high-performance bimetallic catalysts requires careful consideration of the electronic interplay between the nanoparticle and its support, as this synergy provides a powerful tool for tuning catalytic activity. A significant reduction in energy barriers for

CO oxidation on defective graphene suggests that Pt-Au clusters anchored to vacancy sites are promising candidates for low-temperature CO oxidation, since the reaction can occur efficiently even under mild conditions.

3.5 | Simulation of CO Saturation

PtAu nanoclusters are studied here for thermal CO oxidation. These reactions involve a significant amount of CO adsorbed on the surface of the catalyst nanoparticle, which may alter the atomic composition and characteristics of the nanoparticle [81, 82]. Classical molecular dynamics simulations are coupled with machine learning interatomic potentials to study the saturation process of Pt₇Au₇ cluster in free-standing and supported configurations with CO molecules. A graphene layer with monovacancy is used to simulate support for the nanocluster. Monovacancy serves as an anchor for the nanoparticle and preserves migration over the graphene surface during simulation.

Simulation of saturation of Pt₇Au₇ cluster with CO molecules is performed using the classical molecular dynamics method combined with the Moment Tensor Potential [54] (MTP) to describe the interatomic interaction. Different concentrations of CO molecules, ranging from 25 to 400 CO molecules per considered box of 200 × 200 × 200 Å are considered as the environmental surrounding for Pt₇Au₇ cluster. Figure 10a,b shows snapshots from MD simulations of free-standing and supported Pt₇Au₇ cluster with adsorbed CO molecules. Entire simulation boxes containing 25 and 400 CO molecules are shown in Figure S16. The maximum simulation time is 10 ns. Detailed information on the pressure of the CO gas and its concentration at 300 and 500 K used in the simulations is provided in Tables S3, S4 in the Supporting Information. During the simulations of Pt₇Au₇ cluster in a CO environment, the acts of interaction of the individual CO molecule with Pt₇Au₇ cluster are recorded. The dependencies of the number of adsorbed CO molecules on free-standing and supported Pt₇Au₇ cluster on time are shown in Figure 10c,d, respectively. Each step on the curve in the diagram indicates the adsorption or desorption of a CO molecule from a Pt₇Au₇ cluster. Considering the saturation process of CO molecules on free-standing Pt₇Au₇ (see Figure 10c) at the lowest studied concentration of 25 CO molecules per cluster at 300 K, one can see that the adsorption process starts at 0.5 ns and after 10 ns of simulation, the maximum number of adsorbed CO molecules is 12. After 2.5 ns there are 8 CO molecules are adsorbed on the surface of Pt₇Au₇. One can see that there are 7 CO molecules are adsorbed on Pt sites, while 5 of them are adsorbed on Au sites. A similar situation is observed for the supported Pt₇Au₇ cluster, where saturation begins fairly quickly and an equal number of Au and Pt sites are involved in adsorption. However, the total number of adsorbed CO molecules on the supported cluster is lower compared to the free-standing cluster as part of the surface interacts with graphene (Figure 10b). An increase in temperature leads to a different saturation behavior. The saturation curve for 500 K is shown in Figure S19. It is found that in a denser CO environment the number of CO molecules adsorbed on Pt sites is almost 3 times larger than the number adsorbed at Au sites. This behavior can be explained by adsorption-induced transitions in bimetallic nanoparticles [70, 83–87]. This makes the Pt₇Au₇ cluster with Pt atoms on the surface completely covered by CO

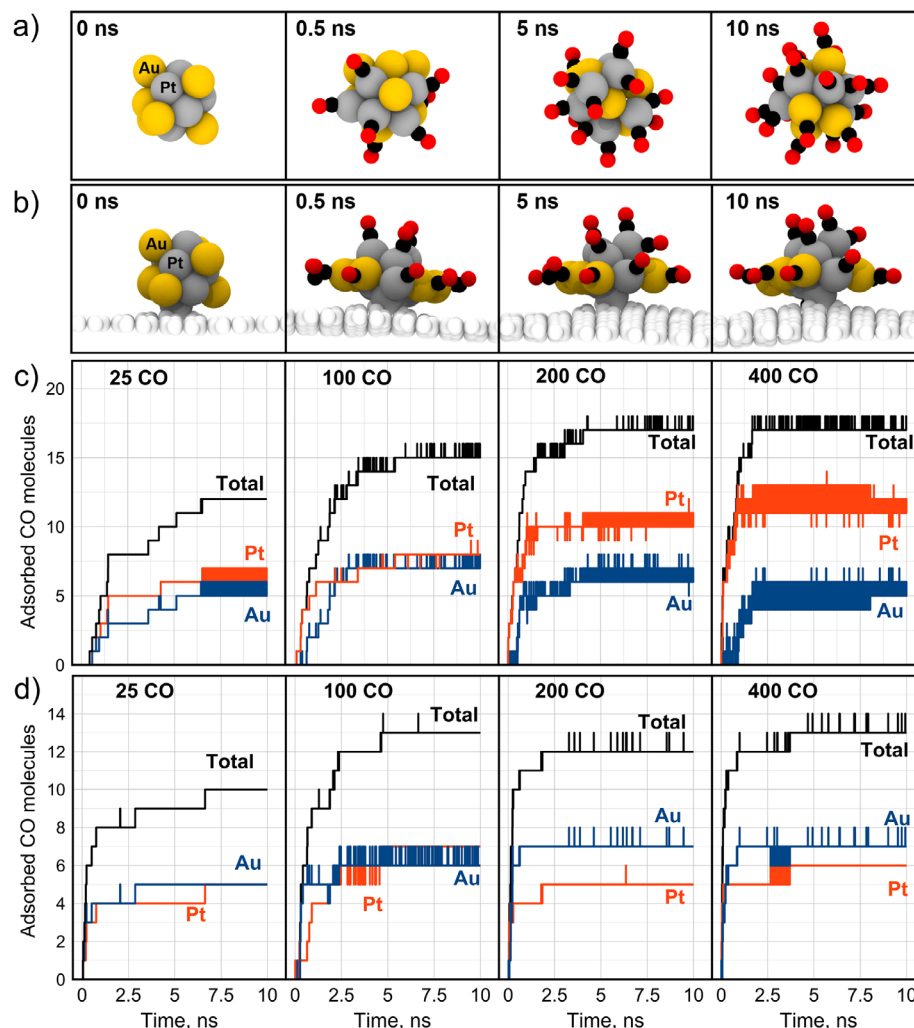


FIGURE 10 | (a) Free-standing and (b) supported Pt_7Au_7 cluster with adsorbed CO molecules at different time points of simulation. Pt atoms are shown by gray color, Au atoms are yellow, carbon is black, oxygen is red, graphene layer is shown by white color. Dependence of the number of adsorbed CO molecules on the (c) free-standing and (d) supported Pt_7Au_7 cluster on the simulation time for different concentration of CO molecules equal to 25, 100, 200, and 400 per Pt_7Au_7 cluster calculated at 300 K. Red color denotes data for CO adsorption on Pt atoms, while blue color is for CO adsorption on Au atoms of Pt_7Au_7 cluster.

molecules energetically more favorable than the initial structure of the cluster without CO, in which all Au atoms are located on the surface.

An increase in CO concentration up to 100 molecules per simulation cell leads to an increased number of adsorbed CO molecules, see Figure 10c. One can clearly see the separation of adsorption sites, i.e. more CO molecules are adsorbed on Pt than on Au. A further increase in the density of CO environments up to 400 CO molecules (at a pressure of 2.072×10^5 Pa), see Figure 10c, leads to an even greater separation due to the partial reconstruction of nanoparticles upon saturation. The separation of adsorption sites is related to the structure reconstruction caused by adsorption-induced transition. It is clearly seen from the Figure S18 that Pt atoms tends to be at the surface of Pt_7Au_7 . This is directly related to the energy efficiency of Pt being located on the surface of the cluster.

Considering the saturation of the supported Pt_7Au_7 cluster (Figure 10d) one can clearly notice a significantly different behav-

ior of adsorption. At low CO pressures (25 and 100 CO molecules), the adsorption behavior is similar to that of a free-standing cluster, but the total number of adsorbed CO molecules is lower, as part of the cluster interacts with graphene (Figure 10b). An increase in the density of CO environments up to 200 and 400 CO molecules leads to rapid saturation with almost equal number of occupied Au and Pt sites, see Figure 10d. This indicates the crucial role of the support in the adsorption process, preserving the reconstruction of the cluster and affecting the entire saturation process. The structural evolution of free-standing Pt_7Au_7 cluster during simulations at different CO pressures and temperatures is shown in Figures S17 and S18.

Based on the molecular dynamics simulations, the saturation of a free-standing Pt_7Au_7 cluster by CO molecules is governed by an adsorption-induced structural transition. This process causes the cluster to restructure, driving Pt atoms to the surface where they become preferred, energetically favorable adsorption sites for CO, especially at higher temperatures and CO concentrations. Consequently, the final structure and CO adsorption profile are a direct

result of this concentration-dependent reconstruction. Graphene support preserves the adsorption-induced reconstruction of the cluster structure leading to different adsorption sites and an entire saturation profile.

4 | Conclusion

In conclusion, this comprehensive computational study reveals that the catalytic properties of small Pt-Au nanoclusters are governed by a complex interplay of their atomic structure, composition, support interaction, and dynamic reconstruction. Using USPEX, we identified stable cluster configurations where Pt atoms preferentially occupy central positions while Au atoms segregate to the surface. High-throughput DFT and ML screening of adsorption energies across both ground-state and metastable isomers demonstrates that adsorption strength is primarily determined by the local composition and coordination environment of the active site, not by the specific global minimum structure. Therefore, for practical adsorption screening, a set of low-energy isomers is as informative as the single most stable cluster, which is particularly important given the experimental difficulty of obtaining monodisperse populations. Analysis of CO oxidation barriers further demonstrates that a monovacancy in graphene alters the kinetic landscape of the Pt₇Au₇ cluster on the substrate. Compared to pristine graphene, the defective substrate reduces the average barrier from 1.34 to 1.03 eV, while simultaneously increasing the number of pathways with a barrier below 1 eV from 12 to 18. The lowest-barrier channels include mixed Au-Pt configurations, demonstrating that Au atoms can become active sites when their electronic structure is modulated by the defective substrate. Thus, the catalytic role of the substrate is two-fold: it stabilizes the nanocluster, preventing migration and coalescence, and electronically rearranges the ensemble of active sites toward more accessible CO oxidation pathways. Molecular dynamics simulations of CO saturation reveal an adsorption-induced structural transition where Pt migrates to the surface, and importantly, the graphene support modifies the saturation kinetics and site occupation compared to the free-standing cluster. This work underscores that rational design of high-performance bimetallic catalysts must consider local site chemistry, electronic synergy with the support, and the realistic isomer distribution in experimental samples, providing new avenues for low-temperature CO oxidation.

Author Contributions

Ilya V. Chepkasov: conceptualization, writing – original draft, investigation, and writing – review & editing. **Alexander G. Kvashnin:** writing – original draft, investigation, writing – review & editing, and supervision. **Alexey S. Galushko, Mikhail N. Khrizanforov, and Valentine P. Ananikov:** writing – original draft. **Viktor S. Baidyshev, Michail M. Lukanov, Aleksandra D. Radina, Mikhail Lazarev, Nikita V. Ter-Oganessian, and Vladimir S. Baturin:** investigation.

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Conflicts of Interest

The authors declare no conflicts of Interest.

Data Availability Statement

All data generated or analyzed during this study, including all structures of nanoclusters predicted during evolutionary search with USPEX code, Moment Tensor Potentials, training set, and Python scripts for creating different configurations of nanoclusters on the graphene support, are available in the following GitHub repository: https://github.com/AlexanderKvashnin/AuPt_uspeX.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: sml175835-sup-0001-SuppMat.pdf.